Concept Review

# Camera Model

## Why Model a Camera?

Cameras are commonplace items today, often a must both from a technological perspective and user's point of view. Cell phones, cars, gaming consoles, drones etc. all deploy this sensor due to its innate ability to provide vision or sight similar to human eyes. Modeling a camera allows us to understand the limitations of projecting a real 3D world onto an image plane, and how features identified in images could be localized and placed back in the real world. While quite a few types of camera sensors exist with corresponding models, this document will focus on the pinhole camera model.

## Camera Extrinsic Properties

A camera's extrinsic definition makes use of a homogeneous transformation matrix to describe a location $\boldsymbol{t_j^i}$ and rotation $\boldsymbol{R_j^i}$ with respect to a desired reference frame. For cameras the extrinsic definition is used to convert a point of light $\overrightarrow{\boldsymbol{P_E}}$ from an environment frame to the camera reference frame. The general form for a homogeneous transformation matrix is,

.

$$\boldsymbol{T_j^i} = \begin{bmatrix} \boldsymbol{R_j^i} & \boldsymbol{t_j^i} \\ \boldsymbol{0} & \boldsymbol{1} \end{bmatrix} \tag{1}$$

The rotation matrix $\boldsymbol{R_j^i}$ can be constructed as the combination of 3 successive rotation about the principal axes $x, y, z$ specified in a particular order. $\boldsymbol{R_j^i}$ defines the relation of frame $j$ with respect to frame $i$. Rotations about each principal axis: (Spong, 2006)

$$R_{x,\phi} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix} \qquad R_{y,\theta} = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \qquad R_{z,\psi} = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation about $x$ axis | Rotation about $y$ axis | Rotation about $z$ axis

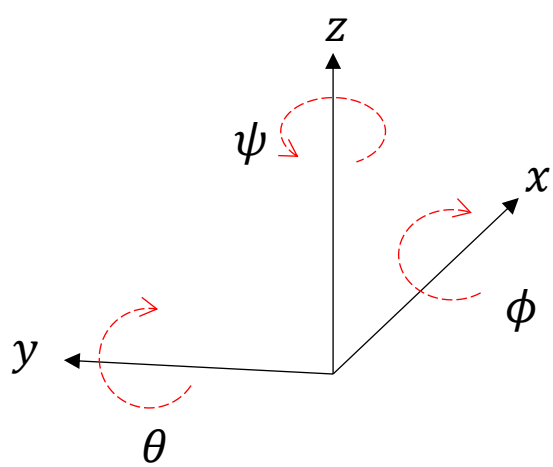


Figure 1: Positive rotation for primary x-y-z axis

With a rotation definition specified the conversion of a point of light $\overrightarrow{\boldsymbol{P_E}}$ from the environment frame $\boldsymbol{E}$ to the camera frame $\boldsymbol{C}$ is,

$$\overrightarrow{P_c} = T_E^c \overrightarrow{P_E} \tag{2}$$

Having defined the point of light in the camera frame the next step is to construct the conversion from a physical point to a pixel in an image.

## Camera Intrinsics

Digital cameras are electromechanical devices which use lenses and optical sensors to transform visible light into a digital image. Prior to using a camera for a visual servoing application we need to calculate the intrinsic parameters which describe how the physical point of reference was transformed into a pixel coordinate. All cameras have the following set of equations:

$$K = \begin{bmatrix} f_x & 0 & C_x \\ 0 & f_y & C_y \\ 0 & 0 & 1 \end{bmatrix} \tag{3}$$

Matrix $K$ describes the principal point and the focal point of a camera in pixel coordinates. This matrix can be calculated using the MathWorks™ Camera Calibration tool and a calibration chessboard. Matrix $K$ also depends on the resolution of the images being read by the camera. Perform a new camera calibration when you want to change the resolution of the camera.

## Pinhole Camera Model

The pinhole camera model is the first step for converting a physical point of light into an image coordinate. Lens distortion is considered later in the camera model formulation. Point of light $\overrightarrow{P_c}$ originates from an optical center called the optical focal point.

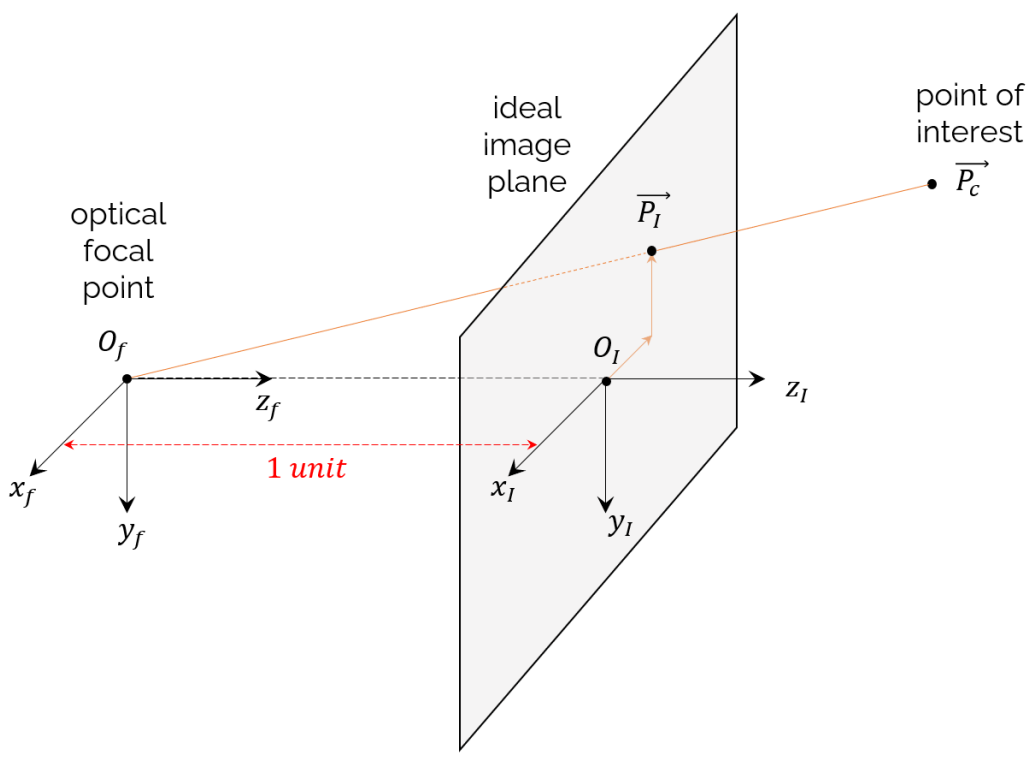


Figure 2: Description of Image Focal point and Ideal image plane. (Waslander, 2021)

Point $\overrightarrow{P_C}$ is used to describe the relationship between the optical axis $O_f x_f y_f z_f$ and the ideal image plane originating at $O_I x_I y_I z_I$. Using image projection and similarity triangles point $\overrightarrow{P_c}$ can be projected onto the ideal image plane using the following relationship,

$$\overrightarrow{P_I} = \begin{bmatrix} Px_I \\ Py_I \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{X}{Z} \\ \frac{Y}{Z} \\ 1 \end{bmatrix} \tag{3}$$

By projection point $\overrightarrow{P_c}$ to the ideal image frame there is a change from $\mathbb{R}^3$to $\mathbb{R}^2$ space. Now in the ideal image plane we consider image distortion due to the type of lens used. The two most common distortion components are,

- Radial distortion ($\boldsymbol{k_{rad}}$)
- Tangential distortion ($\boldsymbol{k_{tan}}$) .

Radial distortion originates at the center of the image plane and depending on the sign of the leading distortion coefficient, can cause an image to warp inwards in a barrel like fashion, or outwards. The combination of both radial and tangential distortion on the ideal image point $\overrightarrow{P_I}$ can be written as,

$$\vec{P}_{Dist} = \begin{bmatrix} Px_{Dist} \\ Py_{Dist} \end{bmatrix} = (1 + k_1 r^2 + k_2 r^4 + K_3 r^6) \begin{bmatrix} Px_I \\ Py_I \end{bmatrix} + \begin{bmatrix} 2\tau_1 Px_I Py_I + \tau_2 (r^2 + 2Px_I^2) \\ 2\tau_2 Px_I Py_I + \tau_1 (r^2 + 2Py_I^2 \end{bmatrix} \tag{4}$$

where $r = \sqrt{Px_I^2 + Py_I^2}$ (Barfoot, 2014). An example of radial distortion is shown in Figure 4.

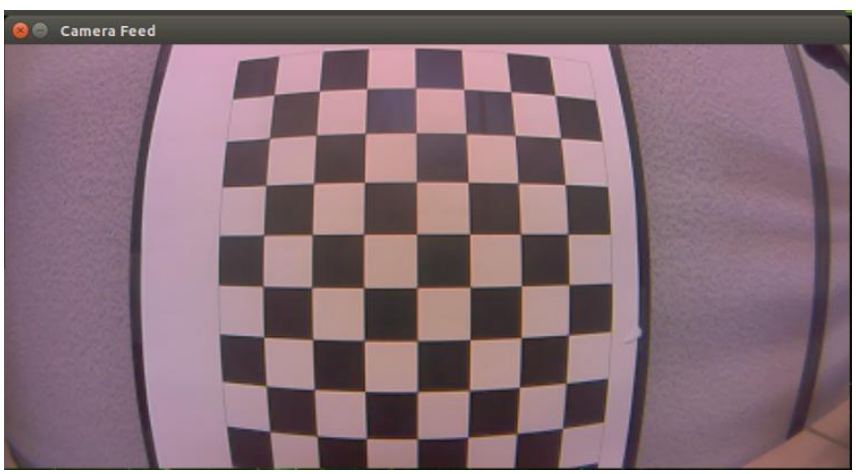


Figure 3: Camera image with barrel distortion.

With the ideal image points modified by the distortion model we now transform from physical coordinates in the ideal image plane to coordinates in the digital image frame. When describing the location of a pixel in an image the following convention is used,

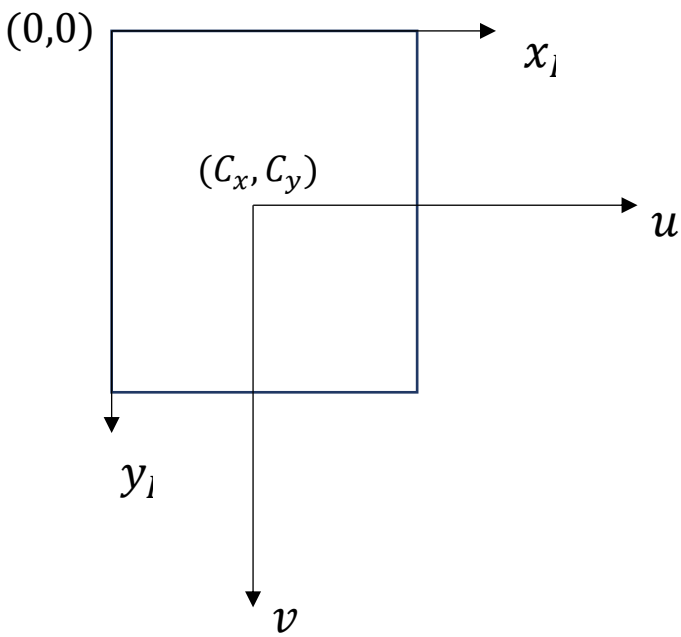


Figure 4: Description of Ideal image to digital image.

With the $u$ axis conciding with $x_I$ and $v$ axis coinciding with $y_I$ as seen in Figure 4. The transformation between the ideal image plane and the camera image plane is defined as,

$$\begin{matrix} u \\ v \end{matrix} = \begin{matrix} f_x P x_{Dist} + C_x \\ f_y P y_{Dist} + C_y \end{matrix} \tag{5}$$

Where $f_x$ and $f_y$ are the focal lengths along the $x$ and $y$ directions. $C_x$ and $C_y$ are centers of the image. All 4 of these parameters are dependent on image resolution. (Barfoot, 2014)

These steps take us from a 3D point to the image frame. Usually, we need to do the reverse and either calculate 3D information based on information from the camera or rectify distorted image data for a give image processing pipeline. A camera calibration process is used to calculate the camera matrix and distortion parameters.

## Summary

Cameras can be modeled using the Camera Obscura (Pinhole camera) model with the inclusion of distortion components. The sequence for transforming a point $\overrightarrow{P_E}$ from the global frame to a pixel coordinate $(u, v)$ is,

Camera Frame Transformation

$$\overrightarrow{P_C} = \begin{bmatrix} \boldsymbol{R} & \boldsymbol{t} \\ 0 & 1 \end{bmatrix} \overrightarrow{P_E}$$

Image Plane projection

$$\overrightarrow{P_I} = \begin{bmatrix} Px_I \\ Py_I \\ 1 \end{bmatrix} = \begin{bmatrix} X/Z \\ Y/Z \\ Z/Z \end{bmatrix} = \frac{\overrightarrow{P_c}}{Z}$$

Image distortion

$$\vec{P}_{Dist} = \begin{bmatrix} Px_{Dist} \\ Py_{Dist} \\ 1 \end{bmatrix} = f(\boldsymbol{k_{rad}}, \boldsymbol{k_{tan}}, \overrightarrow{P_I})$$

Pixel Image coordinates

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & C_x \\ 0 & f_y & C_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} Px_{Dist} \\ Py_{Dist} \\ 1 \end{bmatrix}$$

## Physical Camera Sensor

When purchasing a camera there are always two elements to consider the lens and the camera sensor itself. Lenses allow you to change the effective field of view of a camera. Examples of common lens types:

- Fisheye
- Wide Angle
- Telephoto

Camera sensors define the pixel density, shutter type, power consumption, resolution and framerates. Example common resolutions:

- 1920x1080 (1080p)
- 1280x960
- 1280x720 (720p)
- 640x480

For modern cameras, the lower the image resolution the higher the framerate available for an application. Resolution is just one aspect for camera sensors. The quality of the image is also affected by the camera image array. This is where a camera's megapixel (MP) rating is important. Two cameras can generate the same 640x480 image, but the content of the image will be different between a 1MP and a 5MP camera. The term megapixel is defined at the pixel area of the camera's active sensor array. As an example, the OV5640 is a 5MP camera because the camera sensor has 2592 x 1944-pixel array size or a total area of 5038848 (5MP).

Camera sensors also include the pixel size in µm. The sensor pixel size is useful for calculating the physical focal length of a lens, and the lens field of view. The physical focal length can be estimated as:

$$f_{mm} = f_{pxl}\, p_{xl} \tag{6}$$

Where $f_{mm}$ is the focal length in mm, $f_{pxl}$ is the focal length in pixels and $p_{xl}$is the pixel size in µm.

The field of view for a camera can also be estimated as:

$$FOV_{diag} = \arctan\left(\frac{\sqrt{w^2 + h^2}}{2f_{pxl}}\right) \tag{7}$$

Where $w$ and $h$ are the width and height of the image in pixels.

Due to the lens focal point and the FOV being dependent on the calibration sequence's reported focal length, there can be differences in the manufacturer and the estimated values. These differences can be caused by the image not utilizing the full image sensor (cropping), the pixel size changing as the resolution changes (due to signal processing

techniques such as binning). Always calibrate a camera based on the resolution being used to get an accurate representation of the estimated focal length in mm. Consult the sensor spec sheet to confirm if binning will modify the pixel size as the resolutions change.

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